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The History of the IAA SETI Permanent Committee – 2000 to 2009

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Abstract

This paper is the fourth in a series chronicling the history of the International Academy of Astronautics (IAA) SETI (search for extraterrestrial intelligence) Permanent Committee. The IAA established the SETI Committee in 1974 in response to growing awareness that humanity might one day detect evidence of intelligent extraterrestrial life. The IAA SETI Committee has a wide-ranging mandate to examine all aspects of the search for intelligent extraterrestrial life, including international issues, astrophysical and astronomical observations, biochemistry, exoplanets, complex life and evolution, planetary space missions, SETI search strategies, and the societal, legal, and political ramifications of a confirmed detection. Now in its 52nd year, the IAA SETI Committee is the only international entity that meets annually to organize symposia, select papers for publication, address specialized topics in SETI, and collaborate with other organizations. This paper examines the Committee's activities throughout the 2000s. The decade began with a presentation by IAA SETI Committee Chair Jill Tarter to the United Nations Committee on the Peaceful Uses of Outer Space. The presentation outlined an IAA Position Paper on potential transmissions from Earth in response to a confirmed detection of extraterrestrial intelligent life. Throughout the decade, the Committee worked on updating two key documents (Declaration Protocol and Transmissions Protocol). These documents provide guidance on what to do in the event of a confirmed detection of extraterrestrial intelligence and what to consider if a response from humanity is required upon receiving a signal. The IAA SETI Committee also grappled with the emerging and contentious field of "active SETI." Major SETI search initiatives of the 2000s included the wildly popular SETI@home project, the long-running SERENDIP program, and optical SETI initiatives. In 2007, the Allen Telescope Array became operational, and SETI search programs began focusing on exoplanetary systems as potential sites of extraterrestrial technology. The decade concluded with the launch of the Kepler Space Telescope in 2009, ushering in a new era of exoplanet discovery and innovative technosignature searches.

Keywords: SETI, IAA, extraterrestrial intelligence, exoplanet, protocol

1. Introduction

A paper describing the founding and early years of the IAA SETI Committee* was presented as the John Billingham Cutting Edge Lecture at the 66th IAC in Jerusalem, Israel, in 2015 [1]. A second paper reviewed the activities of the IAA SETI Committee and global SETI developments from 1980 to 1989 and was presented at the 70th IAC in Washington, D.C. in 2019 [2]. The period from 1990 to 1999 was covered in the John Billingham Cutting Edge Lecture at the 74th IAC in Baku, Azerbaijan [3]. This paper chronicles the activities and achievements of the IAA SETI Committee during the decade from 2000 to 2009.

The International Academy of Astronautics (IAA)

* In this paper, the term IAA SETI Committee refers to the IAA SETI Permanent Committee. The IAA SETI Committee was previously known as the IAA CETI Committee until 1987. CETI stands for communication with extraterrestrial intelligence. Today, the IAA SETI Committee is a Permanent Committee under Commission 1 (Space Physical Sciences) of the IAA.

After the Second World War, Germany, Britain, and France organized an international meeting for members of rocket societies and other groups and individuals to discuss non-military space research, space travel, and related topics. The first International Astronautical Congress (IAC) was held in Paris in 1950, and the International Astronautical Federation (IAF) was subsequently established to foster dialogue between scientists and encourage international cooperation for the non-military use of space.

In 1957, humanity entered the space age with the launch of Sputnik. In 1960, the IAF, an organization of member societies, approved the formation of an international organization of distinguished individuals in the aeronautics field, the International Academy of Astronautics (IAA). The International Institute of Space Law (IISL) was established at the same time. The IAA provides a non-governmental forum through which members can contribute to advances in all aspects of the peaceful exploration of space, including space policy, law, economics, and space and society. The IAA and the IISL would, under the umbrella of the IAF, hold joint

symposia and meetings at the annual International Astronautical Congresses. This structure remains in place today.

Introduction of SETI to the IAA

When the IAA was established, the space age was underway, and there was a growing realization that humans might detect extraterrestrial life, including intelligent life.

Project Ozma, the first modern scientific search for radio signals from extraterrestrial intelligence, took place at Green Bank, West Virginia, U.S.A. in 1960. Rudolf Pesěk, a member of the new IAA, valiantly introduced the topic of SETI to his IAA colleagues in 1965. In 1972, the first IAA CETI symposium took place during the 23rd IAC in Vienna, Austria. By 1974, the IAA had established the SETI Committee, with Rudolf Pesěk appointed as its Chair. Of particular importance was the view of the new Committee that papers, talks, and activities should address both the technological and sociological aspects in the search for intelligent extraterrestrial life in the universe. A key priority was encouraging participation by Soviet scientists, who were actively involved in SETI activities at this time [4].

IAA SETI Committee Mandate

The mandate of the IAA SETI Committee is as follows:

1. Examine all aspects of possible future contact with extraterrestrial civilizations, with special reference to international issues and activities involving the underlying science of the study of life in the universe, including:
 - relevant astrophysical and astronomical phenomena
 - the nature and distribution of biogenic elements and compounds
 - the formation of life-bearing planets
 - the origin and evolution of biological systems
 - complex life and intelligence
 - the emergence of technological civilizations
2. The IAA SETI Committee also looks at associated space missions and technology including:
 - planetary science missions designed to search for evidence of extraterrestrial life
 - the astronomical search for extrasolar planets
 - the recovery of possible Martian microfossils on Earth

3. The science and technology of SETI includes:

- the development of search strategies and the underlying intellectual foundation
- the design, development, construction, and operation of telescopes and signal detection systems, studies of their future location in space or on the lunar farside
- rejection of radio frequency interference
- the archiving of SETI data, and analysis of search results.

And if that was not enough, the IAA also states that the IAA SETI Committee will continue to lead discussions on the implications of detecting extraterrestrial signals. Post-detection areas of study include philosophy, historical analogues, anthropology, legal, political, and institutional issues, sociology, psychology, theology, media, and education.

Finally, the IAA SETI Committee organizes annual SETI sessions/symposia as part of the annual International Astronautical Congress (IAC). Traditionally, the first SETI session focuses on technical papers, and the second session is devoted to social sciences and interdisciplinary studies. The Committee selects peer-reviewed papers of interest for publication in the scientific journal *Acta Astronautica*, collaborates with other groups to co-sponsor conferences, convenes annually to discuss SETI matters, and establishes task forces, workgroups, and subcommittees to examine issues of importance to SETI research. The IAA SETI Committee is tasked with preparing for the detection of intelligent extraterrestrial life and considering implications of a potential transmitted response from Earth.

IAA SETI Committee – Early Activities and Accomplishments

One of the inaugural actions undertaken by the new IAA SETI Committee was to advocate for the protection of particular radio frequencies to support the search for extraterrestrial signals. Although the resolution was approved by both the IAA and IAF, the International Telecommunications Union ultimately declined to allocate dedicated spectrum space for SETI activities.

By 1980, Rudolph Pesěk stepped down as the inaugural Chair of the IAA SETI Committee, and John Billingham assumed the role.

As stated in its mandate, the IAA SETI Committee was never just about the science and technology involved in searching for extraterrestrial intelligent life in the universe. It also examines the international, historical, societal, political, legal, institutional,

educational, and media aspects of the search as well as the potential consequences of a confirmed detection. The IAA SETI Committee recognized that SETI researchers could benefit from specific principles or guidelines about what happens if a candidate signal is identified. The verification of a signal, data dissemination, and the public announcement of such a momentous discovery are factors flowing from the initial observation.

To address the need to establish a framework of sorts to consider for any candidate signals, members of the IAA SETI Committee worked diligently throughout the 1980s to produce a document outlining a series of guidelines and principles for individuals or organizations engaged in scientific searches for signs of extraterrestrial intelligence. The Committee worked closely with several experts, most notably from the IISL, to draft a series of guidelines that SETI researchers, mainly radio astronomers, should consider if candidate signals are detected. Most of this work was undertaken by a subcommittee established by the IAA SETI Committee, chaired by member Michael Michaud.

By April 1989, the final seminal document “Declaration of Principles Concerning Activities Following the Detection of Extraterrestrial Intelligence”, or “Declaration Protocol” as it is referred to in this paper, was endorsed by both the IAA and IISL. The Declaration Protocol has no formal government authority and is voluntary; however, it represented a remarkable achievement.[†] A special issue of *Acta Astronautica* featured papers pertaining to the Declaration Protocol [5].

The nine principles listed in the 1989 Declaration Protocol provide guidance on issues such as candidate signal verification, informing other observatories for confirmation, and when to alert authorities, including the Secretary General of the United Nations. The Declaration Protocol states that the news of any confirmed detection of extraterrestrial intelligent life should be disseminated promptly, openly, and widely through scientific channels and public media. The discoverer should have the privilege of making the first public announcement. Data should be collected, and any critical radio frequencies should be protected. The Declaration Protocol states that the IAA SETI Committee, in coordination with Commission 51 of the International Astronomical Union (IAU), will conduct reviews of procedures for detecting extraterrestrial intelligence. The IAA SETI Committee is tasked with maintaining a current list of experts and individuals with

relevant skills to advise on the analysis of a verified signal.

Most notably, Principle 8 in the 1989 Declaration states, “No response to a signal or other evidence of extraterrestrial intelligence should be sent until appropriate international consultations have taken place. The procedures for such consultation will be the subject of a separate agreement, declaration or arrangement.”

By 1996, an IAA Position Paper titled “A Decision Process for Examining the Possibility of Sending Communications to Extraterrestrial Civilizations – A Proposal” was prepared under the direction of Michael Michaud, John Billingham, Ernst Fasan, and others. The IAA Position Paper included two Annexes. Annex I was the 1989 Declaration Protocol, and Annex II was the 1995 Draft Transmissions Protocol. The 1996 IAA Position Paper was sent by the President of the IAA, with the endorsement of the IISL, to the Foreign Ministers or Secretaries of State of the 61 nations that are members of the United Nations Committee for the Peaceful Uses of Outer Space (UNCOPUOS). The goal was to solicit a government willing to introduce the item to the UNCOPUOS agenda, but as is often the case, there was some support for the IAA Position Paper, but no government stepped forward to “go first.”

Extrasolar Planets and Galaxies

The 1990s were marked by the spectacular discovery of extrasolar planets and a growing awareness of the sheer probability of extraterrestrial life through the release of the Hubble Deep Field image in early 1996. This image, taken of a relatively empty area of sky in the constellation Ursa Major, revealed an astonishing array of 3,000 galaxies at various stages of evolution. The surprising 1995 announcement of potential fossils found in a Martian meteorite, and the ensuing controversy and scientific debate was carefully monitored by the IAA SETI Committee. By the end of the 1990s, John Billingham had stepped down as Chair and Jill Tarter took his place.

SETI search projects were conducted throughout the 1990s despite the surprise cancellation of NASA’s SETI program in 1993. Creative thinking and a Herculean effort by the SETI Institute resulted in the transformation of the NASA SETI projects into privately funded Project Phoenix. Meanwhile, Soviet SETI projects faced severe strain during the political and economic upheavals that followed the dissolution of the Soviet Union [6].

[†] The Declaration document was revised in 2010.

2. A New Millennium Marking Forty Years of SETI Searches

A new decade and a new millennium began with optimism as fears of Y2K global chaos dissipated. The global economy was thriving at the top of the dot.com bubble, while people everywhere were gaining access to the new internet and exploring the World Wide Web.

A Presentation to UNCOPUOS

On June 8, 2000, the Chair of UNCOPUOS, invited Jill Tarter, Chair of the IAA SETI Committee, to present the 1996 IAA Position Paper “A Decision Process for Examining the Possibility of Sending Communications to Extraterrestrial Civilizations.” IAA SETI Committee member Ernst Fasan noted that the significant presentation is now in the official record at UNCOPUOS and in the archives of the United Nations. In the event of a signal detection, the United Nations General Assembly can be alerted to the fact that these documents are on file at the United Nations for consideration [7].

Reflecting the robust economy and global optimism, the 51st International Astronautical Congress was held in vibrant Rio de Janeiro, Brazil. During the Congress, more than 700 papers were delivered during 104 sessions.

Excitement and optimism were reflected in the papers presented during the IAA SETI symposium sessions at the Congress. William Welch delivered the Pesek Keynote Lecture, discussing progress on the construction of the Allen Telescope Array. The Pesek Lecture was established in honour of Rudolph Pesek. William Borucki presented on projects aimed at detecting planets orbiting other stars. His Kepler mission proposal, finally approved by NASA in 2000 after years of rejections, would go on to transform humanity’s understanding of the potential for life beyond Earth by identifying thousands of exoplanets. Tom Pierson provided an overview of *SETI 2020: A Roadmap for the Next 20 Years of SETI Research*. Other researchers reported on global SETI initiatives, including Dan Werthimer, who gave updates on the Berkeley programs—SETI@home, SERENDIP, and optical SETI. Andrew Howard presented progress on an all-sky optical SETI survey.

The Rio Scale is Introduced

Ivan Almar and Jill Tarter presented “The Discovery of ETI as a High-Consequence, Low-Probability Event” and introduced the Rio Scale. The scale was designed to quantify the potential significance of discovering extraterrestrial intelligence and to provide a framework that helps prevent misinterpretation or sensationalist

claims about candidate signals or other signs of extraterrestrial technology. To illustrate the concept, they referred to the Richter Scale, a widely recognized tool for measuring earthquake severity that is familiar to both the media and the public. Since 1997, the Torino Scale has similarly quantified the significance of a potential asteroid impact by considering both the damage such an event could cause and the likelihood of collision with Earth. The Rio Scale follows this model, aiming to assess both the credibility of the evidence presented and the potential impact such a discovery would have on humanity.

On September 30, 2000 the IAA SETI Committee convened in Rio de Janeiro. The meeting was chaired by IAA SETI Committee Chair Jill Tarter. Members present included Co-Chairs Claudio Maccone and Tom Pierson, Robert DeBiase, Donald Tarter, Ray Norris, Vladimir Kopal, Roger Malina, Guillermo Lemarchand, Seth Shostak, Carol Oliver, Rosa Maria Ramirez de Arellano, Patricia Sterns, Jill Tarter, and John Rummel. Paul Shuch, a future Co-Chair of the Committee attended as a guest along with Mike Davis and Robin Corbet. Mike Davis, Paul Shuch and Remington Stone from the U.S.A. were recommended as new members along with Ian Morison from the U.K. and Harvey Butcher from the Netherlands.

Claudio Maccone, newly appointed Co-Chair of the IAA SETI Committee, had been collaborating with Jean Heidmann on the Cosmic Study of Lunar Farside Radio Observatory. Sadly, Jean Heidmann, a strong advocate for SETI, passed away in 2000. Claudio Maccone will continue this work.

Ray Norris reported on the informal meeting of the Post-Detection Subcommittee, held in August 1999 during the IAU General Assembly in Manchester, England. Over the past year, the Subcommittee had been restructuring its activities and had established a website, which included both a publicly accessible section and a password-protected area reserved for committee members. Discussions centred on whether the Subcommittee should act only in the event of a confirmed detection or also contribute to preparatory work prior to confirmation in order to assist in the verification process. The consensus favoured earlier involvement, emphasizing the importance of enabling such participation.

The President of the IAA was a special guest at the meeting and described changes on the committee structure within the Academy. He reiterated the support of the IAA for SETI-related activities.

The agenda also included a report from Claudio Maccone on SETI activities in Italy, an update by Guillermo Lemarchand on projects in Argentina, and a presentation by Rosa María Ramírez de Arellano on meetings in Turkey concerning the protection of specific radio frequencies.

NASA Astrobiology Conferences Begin

Committee members also continued their collaborations through research projects and participation in international conferences. The first Astrobiology Science Conference was held at NASA's Ames Research Center in California. This conference, the first in a triennial series, was part of NASA's Astrobiology Program. The program's 1998 Roadmap emphasized an interdisciplinary approach to the search for life beyond Earth and identified a central question: whether life exists elsewhere in the universe. Rather awkwardly for SETI research, while NASA's astrobiology program vigorously investigated the origins and distribution of life in the solar system and beyond, it specifically excluded the search for intelligent extraterrestrial life and the implications of a potential confirmed discovery.

2001 – A Turbulent Year

In April 2001, humanity entered a new era when Dennis Tito became the first space tourist, paying for a flight on a Soyuz spacecraft to the International Space Station.

The 52nd International Astronautical Congress took place in Toulouse, France, under sombre circumstances and with reduced attendance. The ruins of the World Trade Center in New York were still smouldering, and global uncertainty lingered in the aftermath of the attack.

Papers presented in the SETI sessions included an update on the Rio Scale by Ivan Almar and a widely distributed policy paper by Michael Michaud on Ten Decisions that Could Shake the World.

Despite the global turmoil, the 28th annual meeting of the IAA SETI Committee was held on September 29, 2001, with Co-Chair Claudio Maccone stepping in as acting Chair in the absence of Chair Jill Tarter and Co-Chair Tom Pierson. IAA SETI Committee members Robert DeBiase, Roger Malina, Seth Shostak, Carol Oliver, Ivan Almar, Allen Tough, Alexander Ollongren, Douglas Vakoch, Ian Morison, and Ernst Fasan were in attendance. Future IAA SETI Committee members John Elliot, Paulo Musso, and Elizabeth Pioletat attended as guests along with Valerie Ascheri, and M.

Schlumberger. At the meeting, Richard Clar, John Elliot, and Paulo Musso were recommended to the acting Chair as new members.

The meeting in Toulouse was the final gathering of the IAA SETI Committee in its existing form. Clarification was provided regarding the restructuring approved by the IAA Board of Trustees. The IAA SETI Committee would be dissolved and members would transition into the new SETI Permanent Study Group, effective at next year's IAC in Houston.

Claudio Maccone reported on the Cosmic Study of a Lunar Farside Radio Observatory, noting that the scope had been expanded to include protecting the lunar farside from radio frequency interference for the study of frequencies below 15 MHz. Scientists from the Jet Propulsion Laboratory in Pasadena were increasingly involved in this effort, supported by the study's Deputy, Mike Davis.

During the meeting, Seth Shostak observed that the SETI Institute had hosted a website for the IAA SETI Committee for several years. With global interest in SETI and astrobiology on the rise, members discussed the importance of maintaining a dedicated online presence. Paul Shuch offered to reserve the domain iaaseti.org and host the site through the SETI League, beginning immediately.

As in previous years, IAA SETI Committee members were engaged in a wide range of related activities worldwide. Updates were presented on Astrobiology Centres, the SETI League, and national initiatives in Australia and Italy, as well as on signal decoding efforts. Additional reports included progress on the Allen Telescope Array at the SETI Institute, developments in optical SETI programs at Lick Observatory and Parkes Observatory in Australia, and ongoing collaborations on interstellar message composition across multiple disciplines.

The 2002 World Space Conference in Houston

The 53rd International Astronautical Congress (IAC), held in Houston in 2002, was organized jointly with the Committee on Space Research (COSPAR) and branded as the World Space Congress 2002. The event was a large-scale gathering, attracting over 20,000 participants across a diverse range of sessions and activities.

The SETI sessions were well attended and covered diverse topics, including updates on ongoing searches and instrumentation, SETI and astrobiology education, new tools and approaches for detecting extraterrestrial intelligence, the study of animal vocalizations as

analogues for interpreting potential SETI signals, and technical as well as philosophical challenges in deciphering radio transmissions or other extraterrestrial technosignatures. Frank Drake delivered the Pesek Lecture on new approaches in the search for extraterrestrial intelligence.

Carol Oliver and Seth Shostak prepared a press release on the Rio Scale and hosted a media briefing during the Congress.

The newly reorganized IAA SETI Permanent Study Group (formerly the IAA SETI Committee) convened on Saturday, October 12, 2002. Attendees included Chair Jill Tarter, Vice-Chairs Tom Pierson and Claudio Maccone, along with Ernst Fasan, Carol Oliver, Seth Shostak, Robert DeBiase, Allen Tough, Paul Shuch, Steven Dick, Patricia Sterns, Les Tennen, and Ivan Almar. Guests included Elizabeth Back Impallomeni, John Elliot, and Stan Gorgolewski.

The membership of the Permanent Study Group following the IAA reorganization is as follows:

Ivan Almar	Hungary
Valeria Ascheri	Italy
John Billingham	U.S.A.
Elizabeth Back Impallomeni	Italy
Harvey Butcher	Netherlands
Richard Clar	U.S.A.
Michael Davis	U.S.A.
Robert DeBiase	U.S.A.
Steven Dick	U.S.A.
John Elliot	England
Ernst Fasan	Austria
Albert Harrison	U.S.A.
Vladimir Kopal	Czech Republic
Guillermo Lemarchand	Argentina
Claudio Maccone	Italy
Roger Malina	U.S.A.
Michael Michaud	U.S.A.
Stelio Montebugnoli	Italy
Ian Morison	England
Paolo Musso	Italy
Ray Norris	Australia
Alexander Ollongren	Netherlands
Lubos Perek	Czech Republic
Thomas Pierson	U.S.A.
Vladimir Plester	Netherlands
Rosa Ramirez de Allano y Haro	Mexico
John Rummel	U.S.A.
Salvatore Santoli	Italy
Seth Shostak	U.S.A.
Paul Shuch	U.S.A.
Patricia Sterns	U.S.A.

Remington Stone	U.S.A.
Jill Tarter	U.S.A.
Leslie Tennen	U.S.A.
Allen Tough	Canada
Douglas Vakoch	U.S.A.

Membership of the IAA SETI Committee numbered 36, compared to 33 in 1994. However, the composition of the group had shifted, with no members from the former Soviet Union. As noted by Gindilis and Gurvits [6], SETI scientists continued their work in Russia and other post-Soviet states during this period, but the field was severely underfunded.

Jill Tarter announced her resignation as Chair of the Committee and recommended that Seth Shostak succeed her. The IAA Board of Trustees subsequently approved the affiliation of the IAA SETI Committee with Commission 1 – Space Physical Sciences.

The Committee also reviewed updates from its four Subcommittees. Jill Tarter chaired a session of the Post-Detection Science and Technology Subcommittee. It was agreed that this group would become a Task Force under the Committee, tasked with assessing the Rio Scale by applying a range of hypothetical scenarios. Claudio Maccone reported on progress of the Lunar Radio SETI study, which was originally initiated by Jean Heidmann. He was working to expand the project into an IAA Cosmic Study, now focused on the Daedalus crater on the Moon's farside, rather than the Saha crater proposed earlier by Heidmann.

Carol Oliver reported on the activities of the Education and Outreach Subcommittee. Moving forward, the formal group was disbanded, and Carol Oliver was appointed Media and Outreach Officer of the Committee. The Transmissions from Earth Subcommittee, previously chaired by Michael Michaud and later by Ernst Fasan, was also disbanded. In its place, a proposal was submitted for a new Study Group, to be chaired by Jill Tarter with support from Douglas Vakoch and Ernst Fasan. The group's focus would be interstellar message construction, with the aim of briefing UNCOPUOS in 2005.

A New Chair for the IAA SETI Committee – Seth Shostak

In February 2003, the world watched in shock as the Space Shuttle Columbia disintegrated over Texas during reentry, following a successful mission in space. In a more hopeful milestone, on October 15, 2003, China launched its first astronaut aboard the Shenzhou 5, becoming the third nation to achieve independent human spaceflight capability.

The 54th International Astronautical Congress (IAC) was held in Bremen, Germany. At the 32nd IAA Symposium on the Search for Extraterrestrial Intelligence, Co-Chair Claudio Maccone delivered the Pesek Lecture. Reports on active search programs included SETI initiatives in Italy, the targeted optical SETI program at Harvard/Smithsonian and Princeton, and Project Phoenix. Several presentations in the SETI sessions reflected the growing interest in interstellar message construction.

The IAA SETI Committee met on September 30, 2003, under the leadership of its new Chair, Seth Shostak. People in attendance included Seth Shostak (Chair), Tom Pierson (Co-Chair), Claudio Maccone (Co-Chair), Carol Oliver, Ivan Almar, Alexander Ollongren, John Elliot, Paulo Musso, Stelio Montebugnoli, Bruce Dorminey, Roger Malina, Mark Lupisella, Richard Carrigan, Ernst Fasan, Andrew Howard, Jason Gallicchio, and Paul Shuch. The Committee expressed its appreciation for Jill Tarter's many years of leadership and dedication. At the same meeting, Tom Pierson stepped down as Co-Chair, and Carol Oliver was nominated and approved as the new Co-Chair. Don Tarter, Jader Monari, Andrew Howard, and Jason Gallicchio were nominated and approved as members of the Committee.

One of Shostak's first initiatives as Chair was to reestablish the Transmissions from Earth Subcommittee. He emphasized the high level of international attention this work continued to receive, as well as the need to update the existing Protocol documents and pursue related discussions with UNCOPUOS. He proposed Michael Michaud as Chair of the Subcommittee, with Ernst Fasan and Douglas Vakoch as Co-Chairs.

The intent in 2004 was for the IAA SETI Committee to update the 1989 Position Paper, including Annex I (updated Declaration Protocol) and Annex II (updated Transmissions Protocol). Upon approval, the revised document would be submitted to the IAA Board of Trustees and the International Institute of Space Law (IISL) for review and approval. The IAA would then request that UNCOPUOS replace the original version on file, presented by Jill Tarter in 2000, with the updated Position Paper.

Planetary exploration in 2004 was highlighted by NASA's Mars Exploration Rovers, Spirit and Opportunity, as well as the arrival of the Cassini spacecraft in orbit around Saturn. The era of privately funded human spaceflight also began with the 24-minute flight of SpaceShipOne, which crossed the 100 km boundary marking outer space.

The 55th International Astronautical Congress (IAC) was held in 2004 in beautiful Vancouver, British Columbia, Canada. Two SETI sessions reflected a wide range of topics. To accommodate the unusually high number of papers submitted, the Pesek Lecture was deferred. Highlights included an overview of the top 200 candidate signals from 25 years of Berkeley SETI research, updates on the construction of the Allen Telescope Array, and presentations on SETI-Italia initiatives. Other contributions addressed themes derived from the Drake Equation, including the evolutionary neurobiology of intelligence and reflections on the impact and implications of discovering extraterrestrial life.

The 31st meeting of the IAA SETI Committee was held with Chair Seth Shostak and Co-Chair Claudio Maccone presiding. Members in attendance included Michael Michaud, Robert DeBiase, Allen Tough, Paul Shuch, Steven Dick, Paulo Musso, and Alexander Ollongren. Guests included Cathy Rand, Lori Walton, Yvan Dutil, and Richard Carrigan. New members elected to the Committee were Yvan Dutil, Richard Carrigan, and Lori Walton. Seth Shostak reported the recent resignation of Carol Oliver as Co-Chair of the IAA SETI Committee. Paul Shuch was nominated and confirmed as the new Co-Chair.

The Post-Detection Subcommittee reported the development of a wiki-based website to serve as a communications tool. Claudio Maccone, Chair of the Lunar Farside Subcommittee, noted that a new Cosmic Study was being proposed to expand the scope of work to include the entire Moon, rather than only the farside.

Transmissions from Earth – Active SETI

By the time the IAA SETI Committee met in Vancouver, the issue of "Active SETI" had generated increasing controversy. The controversy was not new: debate over disclosing the existence and location of Earth to unknown entities dated back to the first intentional message that was transmitted into space in 1962, when Soviet radio astronomers at the Evpatoria Planetary Radar in Crimea, Ukraine, sent a Morse code signal to Venus as an equipment test. In 1974, Frank Drake and colleagues designed a message transmitted from the Arecibo radio telescope to the M13 globular cluster, more than 25,000 light-years away. Similarly, the plaques and records placed on the Pioneer and Voyager spacecraft provided information identifying the location of the solar system.

By the early 2000s, the Evpatoria facility was once again being used to transmit powerful interstellar

messages. In 1999, Cosmic Call I was sent, followed by the Teen Age Message in 2001, Cosmic Call II in 2003, and Message from Earth in 2008. In the same year, NASA arranged to transmit the Beatles' song Across the Universe toward the star Polaris from the Deep Space Communications Complex in Madrid, Spain. In 2009, NASA supported the Australian project, Hello from Earth, which transmitted a collection of human messages to Gliese 581d, an exoplanet orbiting within the habitable zone of its parent star, located only 20 light-years away.

Several members of the IAA SETI Committee were directly involved in designing and transmitting these messages. Others were studying the implications of what these messages, spanning space and time, might convey about humanity to unknown extraterrestrials, whether biological or machine. Although Active SETI transmissions remained sporadic, their frequency was steadily increasing as people and businesses were intrigued with the idea of sending messages to interstellar space.

Updating the Protocol Documents

Controversy over Active SETI spilled into the hard work of the Transmissions from Earth Subcommittee, which had been tasked with updating both the Declaration Protocol and the Transmissions Protocol. The central point of contention was whether some process of international consultation should occur before transmitting a message, and whether this consultation would apply to messages transmitted intermittently in Active SETI projects or only in response to a confirmed detection of extraterrestrial intelligence.

The debate largely arose from subtle differences in wording between the two documents.

The Declaration Protocol, approved in 1989 and presented to UNCOPUOS in 2000 as Annex I of the IAA Position Paper, states in Principle 7: *"No transmission to extraterrestrial intelligence in response to a signal or other evidence of extraterrestrial intelligence should be sent until appropriate international consultations have taken place."* This wording applies only to transmissions sent in response to a detection and does not explicitly address Active SETI projects. Consequently, the implication is that no international consultation is required for transmissions initiated from Earth in the absence of a confirmed signal.

The draft Transmissions Protocol, developed several years later and presented to UNCOPUOS in 2000 as

Annex II, adopts broader language. Its Principle 7 states: *"No communication to extraterrestrial intelligence should be sent by any State until appropriate international consultations have taken place. States should not cooperate with attempts to communicate with extraterrestrial intelligence that do not conform to the principles of this Declaration."* This wording implies that Active SETI transmissions should not proceed without international consultation.

By 2004, the Subcommittee had streamlined the language and attempted to align the two Protocols. The most controversial change by the Subcommittee was to Principle 7 of the Declaration Protocol, which was modified to read: *"No transmissions should be sent to extraterrestrial intelligence until appropriate international consultations have taken place."* This revision now mirrored the intent of the Transmissions Protocol.

The Subcommittee included Michael Michaud and John Billingham, as well as invited experts from multiple disciplines. Between the Subcommittee and the IAA SETI Committee, two distinct perspectives emerged.

Those opposed to prior consultations for Active SETI projects argued that requiring consultation before any transmission infringed upon free speech and individual rights. They noted that the documents were guidelines rather than binding law and that the IAA had no authority to enforce compliance with them. Earth had already been leaking electromagnetic signals for decades. From this view, active transmissions offered an opportunity to advance scientific knowledge in areas such as target selection and message design.

The other group, most notably the Transmissions from Earth Subcommittee member David Brin, insisted that no transmissions should be sent without prior international consultation. They argued that the risks to humanity were too great and pointed to longstanding concerns about revealing Earth's location, as in the Pioneer plaques, Voyager records, and the 1974 Arecibo message. They emphasized that deliberate, high-powered transmissions differed significantly from background electromagnetic leakage and could provoke a response from extraterrestrial civilizations of unknown intentions. There were concerns that Active SETI activities might create confusion with traditional passive searches, potentially undermining and diverting scarce funding.

2005 – A Busy Year for Global SETI Searches

The year began with the landing of the Huygens probe on Titan, Saturn's largest moon, and humanity was able to get a first look at mountains and river channels on this distant moon. Later that year, the Deep Impact space probe slammed into a comet, enabling scientists to examine the impact debris and identify water ice within the comet. A highlight of 2005 was the launch of the space shuttle Discovery on July 26, 2005, two and a half years after the loss of the space shuttle Columbia.

For SETI searches, 2005 was a period of significant activity. At Berkeley, the SERENDIP project continued, and the highly successful SETI@home program transitioned at the end of the year to the BOINC (Berkeley Open Infrastructure for Network Computing) platform, which supported distributed computing for multiple scientific projects. Since its launch in 1999, SETI@home has achieved global visibility, with its iconic screensaver downloaded by millions of users to both personal and workplace computers. At the same time, astronomers were evaluating exoplanets for habitability, while optical SETI research advanced at Lick Observatory.

Prior to the IAA SETI Committee meeting in Fukuoka, Japan, IAA SETI Committee Chair Seth Shostak circulated proposed draft revisions to the Protocol documents, drafted by the Transmissions from Earth subcommittee. The package included a detailed explanation by John Billingham of the proposed changes. Members were asked for their comments and whether they would support the proposed amendments, particularly the non-enforceable requirement for international consultation before sending any transmissions from Earth.

During the SETI sessions, Ivan Almar and Paul Shuch introduced the San Marino Scale. The scale attempts to quantify the potential hazard of Active SETI projects.

The IAA SETI Committee met on Monday, October 17, 2005, and was convened by Chair Seth Shostak. Co-Chairs Claudio Maccone and Paul Shuch were in attendance along with members Richard Clar, Robert DeBiase, John Elliot, Carol Oliver, John Rummel, Douglas Vakoch and Lori Walton. Kathryn Denning was nominated and approved as a new member.

At the meeting, Seth Shostak reported on member Allen Tough's generous offer to help fund an annual stipend for the John Billingham Cutting Edge Lecture to take place at the start of the SETI II session at each IAC. The Committee agreed that the Chair of the IAA SETI Committee would solicit nominations for the

lecture, to be followed by a membership vote. Members also thanked Paul Shuch and the SETI League for continuing to host the IAA SETI Committee website and recognized Shuch for his service as webmaster.

Subcommittee reports included an update from Claudio Maccone on efforts to protect the Daedalus crater on the lunar farside as a site of scientific value. Carol Oliver provided a report on the Post-Detection Subcommittee, which was developing a 24/7 alert system, a dedicated wiki, and a more clearly defined post-detection policy. Pierre Molette, the editor of *Acta Astronautica* for special issues, was a guest at the meeting. He provided an overview of procedures for publishing SETI papers.

A Key Vote on Transmissions from Earth and Prior Consultation

At the Fukuoka meeting, Seth Shostak reported on the comments and votes back from members regarding the two updated Protocol documents. A motion was passed urging the Chair of the Transmissions from Earth Subcommittee to redraft the Transmissions Protocol document to reflect that no transmissions should be made in response to the detection of extraterrestrial intelligence without international consultation. The Transmissions from Earth Subcommittee had proposed that the wording be broader, covering Active SETI transmissions, not just in response to a confirmed signal.

He also reported that several people commented they would support a separate Active SETI conference or session.

City of Arts and Sciences – Valencia, Spain in 2006

The 57th IAC took place at the futuristic City of Arts and Sciences conference venue in Valencia, Spain. Attendance at the IAC was high, reflecting a robust global economy.

Both SETI sessions were well attended. Claudio Maccone presented the Pesek Lecture and spoke on SETI, extrasolar planets, and interstellar flight. The inaugural John Billingham Cutting Edge Lecture at the start of SETI Session II was presented by Steven Dick. He examined the Drake equation and the concept of technical civilizations evolving to a post-biological existence, along with the cultural impact of striving to improve intelligence.

At the IAA SETI Committee annual meeting, Seth Shostak (Chair), Co-Chairs Claudio Maccone and Paul Shuch were present, along with members Ivan Almar, Linda Billings, Richard Carrigan, Robert DeBiase,

Kathryn Denning, Steven Dick, John Elliot, Andrew Howard, Earnst Fasan, Curtis Mead, Michael Michaud, Jader Monari, Stelio Montebugnoli, Paolo Musso, Ray Norris, Carol Oliver, Alexander Ollongren, Salvatore Pluchino, Doug Vakoch, and Lori Walton.

The SETI league will continue to host the IAA SETI Committee website, and all members thanked Paul Shuch for his outstanding service as the webmaster for the site.

Claudio Maccone reviewed the issue of protection for the Daedalus crater on the far side of the moon. The members present all agreed that protecting the crater is important.

Lingering Discontent with the Wording of the Transmissions Protocol document

Michael Michaud provided the IAA SETI Committee with an overview of the persisting issue regarding the re-drafting of the Transmissions Protocol. He emphasized the history and background behind the Transmissions from Earth Subcommittee's recommended wording. He urged the IAA SETI Committee to reconsider its vote to redraft the Subcommittee's recommended wording. Ultimately, it was decided that a small committee would undertake the task of revising the Transmissions Protocol to its original language.

The IAA SETI Committee engaged in a robust discussion regarding the idea of hosting an "Active SETI" conference and related issues. There was discussion about the possibility of a separate group within the IAA that focuses on active SETI. There was concern about IAA support for such a venture and fractionation of the existing IAA SETI Committee.

Immediately after the Valencia IAC, an editorial was published in Nature [8] regarding deliberate signals sent into space. The article noted that the IAA SETI Committee voted against establishing any processes for reviewing the style or content of messages sent to space during Active SETI transmissions. The editorial cautions that there is a small risk associated with deliberate transmissions that should be taken seriously.

2007 – An Exoplanet in the Habitable Zone

In 2007, the discovery of Gliese 581c was announced. The identification of an exoplanet slightly larger than Earth and in the habitable zone of a red dwarf star generated considerable excitement. Also in 2007, the introduction of the first iPhone marked the

beginning of a fully mobile internet, social media, and content sharing.

John Billingham, who played such an influential role in the early days of SETI and held Co-Chair and Chair positions in the IAA SETI Committee, submitted a letter of resignation in September 2006 to Chair Seth Shostak and the other Committee members. He had tirelessly advocated for and ensured that the IAA SETI Committee examine all aspects of the scientific search for extraterrestrial intelligent life in the universe, and personally advocated for studying the social and cultural impact on humanity of a confirmed detection. Michael Michaud also submitted his resignation at this time, after decades of diligent work leading efforts to draft the original Declaration Protocol, the Transmissions Protocol, and then the updated versions.

At the 58th IAC in Hyderabad, India, Seth Shostak presented an update on the Allen Telescope Array and the galactic center SETI search. The Allen Telescope Array, operated by the University of California, Berkeley and the SETI Institute, began operations with 42 of a planned 350 radio dishes, making it the largest telescope devoted to SETI searches. Govind Swarup presented the Pesek Lecture in the IAA SETI Symposium and Kathryn Denning presented the Billingham Cutting Edge Lecture. The SETI sessions were well attended.

The annual IAA SETI Committee meeting convened with Chair Seth Shostak, Co-Chairs Claudio Maccone and Paul Shuch and members Robert DeBiase, Kathryn Denning, and Douglas Vakoch along with guest Alexander Zaitsev.

Claudio Maccone reported that the latest version of the Transmissions Protocol document was presented to the IAA in March 2007. Seth Shostak stated that the Transmissions Protocols had been presented in Hyderabad on September 23 to Commission I. The only significant point of discussion was that the phrase "international consultation" remained vague, and it was suggested that the decision-making body should be more clearly specified.

2008 – A Global Economic Collapse

The 59th International Astronautical Congress took place in Glasgow, Scotland, from September 29 to October 3, 2008, against the backdrop of the most severe global financial crisis since the Great Depression. During this period, Lehman Brothers filed for bankruptcy in the U.S.A., Iceland's banking system collapsed, and the UK government nationalized Northern Rock. Despite the gloomy financial turmoil,

construction of the International Space Station continued, the New Horizons probe was launched to Pluto, and SpaceX successfully launched its Falcon 1 rocket into orbit for the first time. By 2008, the popular SETI@home project had signed up more than 5 million volunteers to utilize their desktop computers to process data from the Arecibo radio telescope in Puerto Rico.

Despite the economic turmoil, it was business as usual at the IAC. William Welch, as the Pesek Lecturer, presented a talk on the Allen Telescope array and Ivan Almar presented the John Billingham Cutting Edge Lecture on SETI terminology. The annual IAA SETI Committee meeting was well attended with 15 members present. In attendance were Ivan Almar, Richard Clar, Carol Oliver, Robert DeBiase, Kathryn Denning, John Elliot, Ernst Fasan, Al Harrison, Claudio Maccone, Stelio Montbugnoli, Alexander Ollongren, Seth Shostak, Paul Shuch, Patricia Sterns, and Lori Walton.

The first order of business at the IAA SETI Committee meeting was to discuss the success of the inaugural IAA Symposium on Searching for Life Signatures from September 22-26, 2008, at the iconic UNESCO building in Paris, France. Organized prior to the IAC in Glasgow, Claudio Maccone, working with the IAA, was the driving force behind the symposium with help from IAA SETI Committee members Kathryn Denning and Paul Shuch. Several SETI Committee members attended and presented at the Searching for Life Signatures Symposium. The success of the IAA Signature for Life Symposium generated a fruitful discussion at the IAA SETI Committee meeting on future symposia and coordination with the annual International Astronautical Congresses.

The IAA SETI Committee also discussed the February 2008 meeting of the Post-Detection Subcommittee, hosted in Arizona by Chair Paul Davies. Subcommittee members Les Tennant, Patricia Sterns, and others emphasized the importance of allowing all Post-Detection Subcommittee members to review and comment on proposed revisions to the Declaration Protocol document.

IAA SETI Committee member John Elliot was acknowledged for all his dedicated efforts towards the publication of SETI papers from past IACs in *Acta Astronautica*.

The aftermath of the global economic crisis and resulting Great Recession placed further funding pressures on SETI search projects. A highlight was the launch on March 7 2009 of the highly anticipated Kepler spacecraft. Over its mission, Kepler confirmed

more than 2600 exoplanets, including several Earth-sized planets located within habitable zones.

The Decade Draws to a Close – Daejeon, South Korea

The 60th International Astronautical Congress took place at Daejeon, South Korea. The 38th IAA SETI Symposium at the IAC once again featured two sessions on SETI. The 1999 Pesek Lecture was presented by Myung-Hyun Rhee. The lecture focused on South Korea's new SETI initiatives, which included a new all-sky SETI survey, educational programs, development of a distributed computing platform called Korea@Home, and various workshops and projects for writers, artists, and the general public. The array of projects reflected the multidisciplinary nature of the search for extraterrestrial intelligent life in the universe. The Billingham Cutting Edge Lecture was presented by James Gardner, who spoke on the relevance of cosmological models to SETI.

The 36th meeting of the IAA SETI Permanent Committee, the last of the decade, was held on October 13, 2009. Chair Seth Shostak and Co-Chairs Claudio Maccone and Paul Shuch were joined by member Douglas Vakoch. Paul Davies and Curtis Mead were nominated and approved as new members.

On the agenda was an update on the IAA SETI Committee's continued work on revising the Post-Detection Protocols. IAA SETI Committee members Les Tennant and Patricia Sterns suggested edits for the revised Protocol documents, and the final draft was to be circulated to the membership for a vote to accept or reject.

Collaboration with other organizations remained a key aspect of the IAA SETI Committee's work, through formal partnerships and through the efforts of individual members. The IAA SETI Committee endorsed the Royal Society SETI meeting for October 4-6, 2010, in Milton Keynes, U.K., as part of its ongoing commitment to international cooperation.

The 2000s marked a period of transition for the IAA SETI Committee, which was reorganized as the IAA SETI Permanent Study Group. During this decade, the Committee continued to facilitate international cooperation on issues of importance to the scientific search for extraterrestrial intelligent life in the universe. Of particular significance was the discovery of an abundance of extrasolar planets, which led to a greater understanding of the nature of other solar systems and their habitable zones. The IAA SETI Committee worked diligently on updating the Declaration and Transmissions Protocol documents and weathered a

contentious policy debate concerning Earth-based transmissions. Despite funding challenges, SETI research and searches expanded through distributed computing initiatives, new observatories, and the growing number of identified exoplanets. By 2009, the field was positioned for further growth, informed by Kepler data and debates over technosignatures, active SETI, and post-detection protocols. The IAA's sustained support ensured that SETI retained a unique international forum at the intersection of science, policy, and society.

The IAA and Decades of Support for SETI

The IAA SETI Committee, with the support of the IAA, remains unique as the sole long-term non-governmental international body that meets annually to present papers, deliver talks, and examine all aspects of the search for intelligent extraterrestrial life. Its work extends beyond technical issues to include the legal, policy, and societal implications of a confirmed detection of evidence of intelligent extraterrestrial life.

With the proliferation of new astrobiology conferences, workshops, and related events in the 2000s, the question of whether the IAA SETI Committee “belonged” within the IAA resurfaced throughout the 2000s, as it had many times before. A persistent concern was the high cost of registration and travel required to attend the International Astronautical Congress, which posed a barrier to participation. The two sessions making up the SETI symposium and the annual meeting of the IAA SETI Committee were small components of the much larger 5-day Congress. Jean-Michel Contant, Secretary General of the IAA, acknowledged these challenges and supported the committee's suggestion of a two-tier fee schedule, based on the number of days attended, as a means of increasing participation.

Despite these difficulties, the IAA SETI Committee recognized the Academy's decades-long advocacy and support of SETI. The IAA provides access to a broad range of multidisciplinary expertise, including art and culture, and maintains a close working relationship with the International Institute of Space Law (IISL). The SETI symposium consistently attracts a strong selection of papers, and attendance at both the symposium sessions and committee meetings is typically robust. Most significantly, the IAA holds recognized standing in policy matters with the United Nations, ensuring a communication channel for SETI-related developments, including the potential confirmation of evidence of extraterrestrial technology.

3. The IAA SETI Committee in 2010 and Beyond

The challenging funding conditions faced by SETI researchers after the 2008 global economic downturn only deepened in the following years. By 2011, even the Allen Telescope Array had gone into hibernation. At the same time, the rapid expansion of exoplanet discoveries by Kepler, TESS, and other instruments enabled astrobiologists and SETI scientists to refine the parameters in the Drake Equation and better assess the likelihood of detecting evidence of intelligent extraterrestrial life. Funding for SETI research continued to be largely through private donations and support from academic institutions.

A transformative change occurred on July 20, 2015, with the announcement of the Breakthrough Listen initiative. This unprecedented \$100 million commitment of private funding[‡] sparked a surge of SETI projects, research programs, and student training opportunities. In 2018, the U.S. Congress reversed its longstanding opposition to funding searches for extraterrestrial intelligence and began supporting research on technosignatures. [9].[§] Financial stability for SETI research was further strengthened in 2023, when the SETI Institute received a \$200 million bequest, creating a secure foundation for expanding search efforts. By 2025, the landscape of SETI research had been reshaped by these infusions of significant and stable funding.

Advances in computing power and the application of artificial intelligence further transformed the field, dramatically accelerating data processing and significantly broadening the scope of SETI observations. During this period, the IAA SETI Committee, with the support of the IAA, continues to work on cross-disciplinary aspects of SETI, updating the two Protocol documents to keep pace with technological advances and societal expectations, contributes to *Acta Astronautica*, and monitors SETI search efforts worldwide.

ACKNOWLEDGEMENTS

I thank the former IAA SETI Committee Chair Claudio Maccone for suggesting this topic many years ago. Most sadly, Claudio passed away recently, but he would have enjoyed this history-related topic. I would

[‡] In 2025 funding was renewed for another ten years

[§] Technosignatures represent any sign of technology that infers the existence of intelligent life elsewhere in the universe, including familiar objects of searches for extraterrestrial intelligence such as narrow-band radio signals or pulsed lasers. The term SETI (Search for Extraterrestrial Intelligence) is often used synonymously with the search for technosignatures [9].

also like to acknowledge the efforts of the late Stephane Dumas in cataloging SETI publications and Ivan Almar for his contribution of historical papers related to SETI. Jill Tarter and Chris Neller supported the research of IAA SETI Committee-related documents held at the SETI Institute.

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